Name

Mathematics Advanced Year 11

Circles, Semicircles, Square Root Function

Book 1

Version: 261021

Feedback:
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Contents

Syllabus Content 2
Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MAV-11-02 uses functions and relations to model, analyse and solve problems

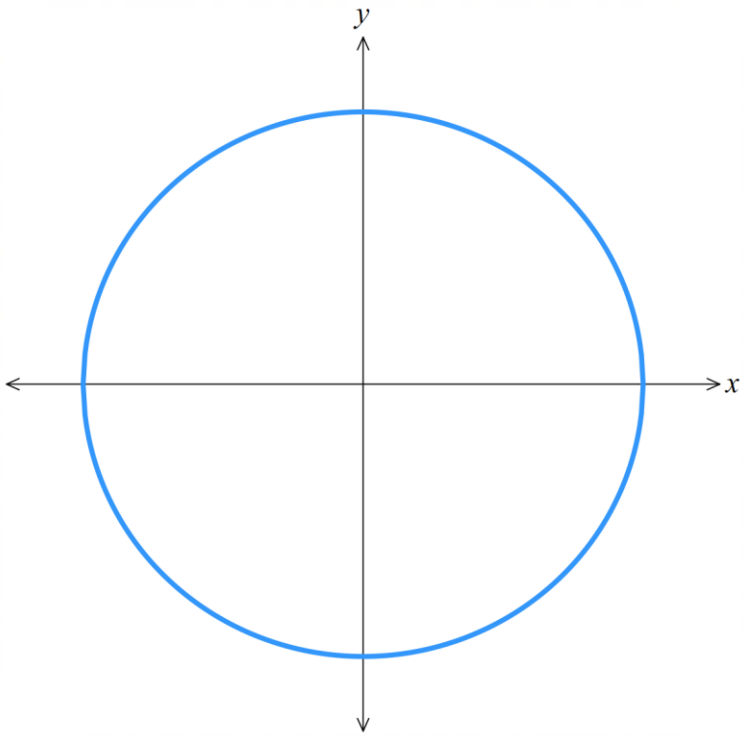
Circles and semicircles

- Derive the equation of a circle of radius r with centre at the origin by considering Pythagoras' theorem
- Graph circles of the form $x^2 + y^2 = r^2$ from their equations
- Determine the equation of a circle of the form $x^2 + y^2 = r^2$ given its graph
- Identify and graph the semicircles $y = \sqrt{r^2 - x^2}$, $y = -\sqrt{r^2 - x^2}$, $x = \sqrt{r^2 - y^2}$ and $x = -\sqrt{r^2 - y^2}$

Circles at the Origin

Investigation Equation of a Circle

Consider the following circle with radius r centred at the origin $(0,0)$



1. Choose a point on the circumference. Label this $P(x, y)$

2. Draw a line segment from the centre to the point P.
   What is the length of this line and what is it called? ……

3. Construct a right-angled triangle using the radius as the hypotenuse.
   Label the lengths of the sides.

4. Write an equation relating the three sides of the right-angled triangle.

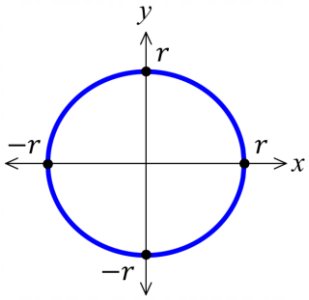
   ……

   This is the equation of a circle.

ⓘ Equation of a Circle Centred at the Origin

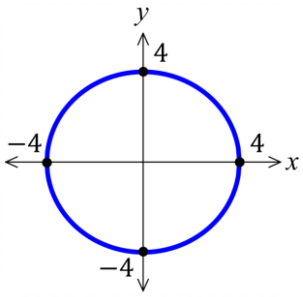
$$x^2 + y^2 = r^2$$

↑ radius r

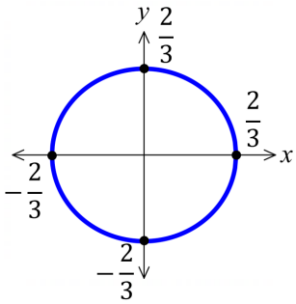


Sketch the graph of a circle centred at the origin

Sketch $x^2 + y^2 = 16$

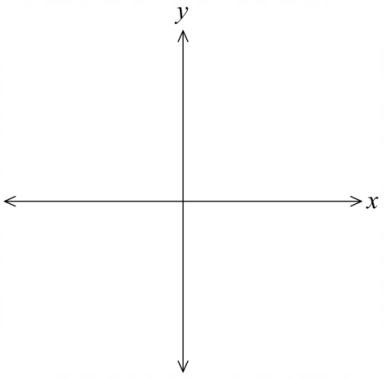


Sketch $x^2 + y^2 = \frac{4}{9}$

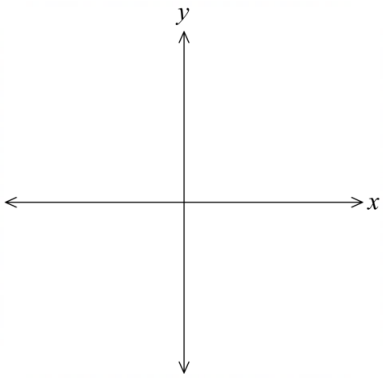


Sketch these circles

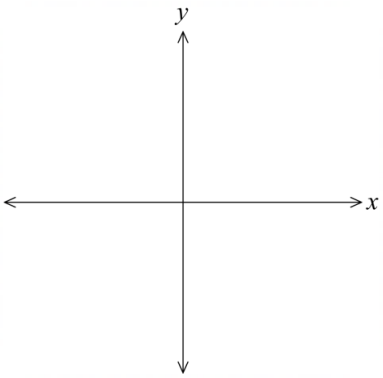
a $x^2 + y^2 = 36$



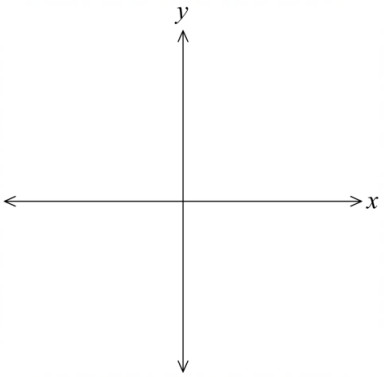
b $x^2 + y^2 = 37$



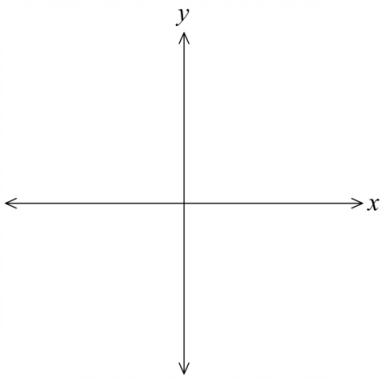
c $x^2 + y^2 = \frac{1}{4}$



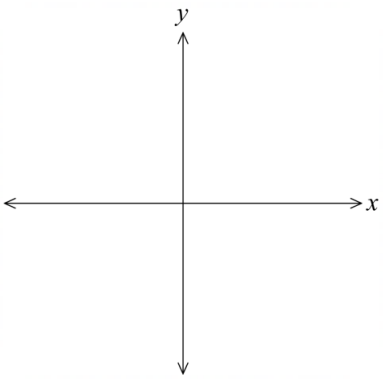
d $x^2 + y^2 = \frac{9}{4}$



e $x^2 + y^2 = 144$



f $x^2 + y^2 = 7^2$



1 Identify the radius of the circles with these equations:

a $x^2 + y^2 = 36$

6

b $x^2 + y^2 = 81$

9

c $x^2 + y^2 = 144$

12

d $x^2 + y^2 = 5$

$\sqrt{5}$

e $x^2 + y^2 = 12$

$2\sqrt{3}$

f $x^2 + y^2 = \frac{1}{4}$

$\frac{1}{2}$

2 Write the equation of a circle with centre at the origin and radius of:

a 4

$x^2 + y^2 = 16$

b 7

$x^2 + y^2 = 49$

c 100

$x^2 + y^2 = 10000$

d $\sqrt{6}$

$x^2 + y^2 = 6$

e $\sqrt{10}$

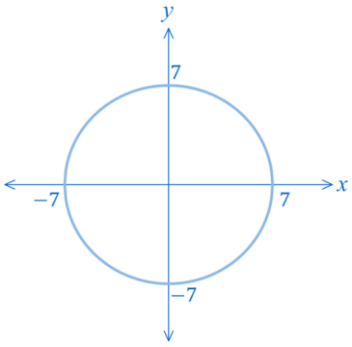
$x^2 + y^2 = 10$

f 1.1

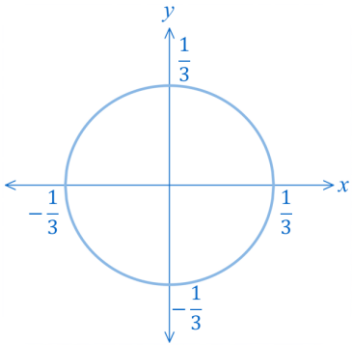
$x^2 + y^2 = 1.21$

3 Sketch these circles showing the centre and all intercepts.

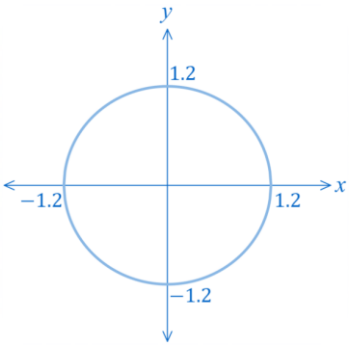
a $x^2 + y^2 = 49$



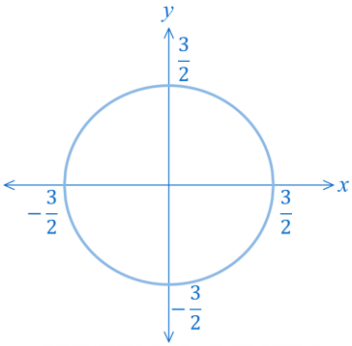
b $x^2 + y^2 = \frac{1}{9}$



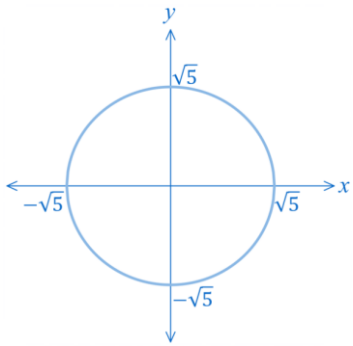
c $x^2 + y^2 = 1.44$



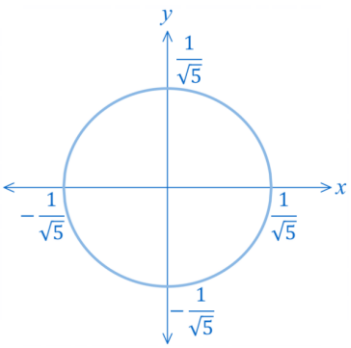
d $x^2 + y^2 = \frac{9}{4}$



e $x^2 + y^2 = 5$

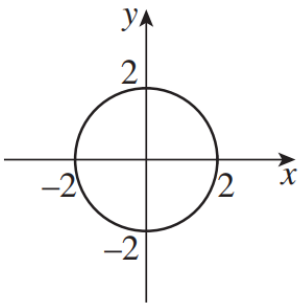


f $x^2 + y^2 = \frac{1}{5}$



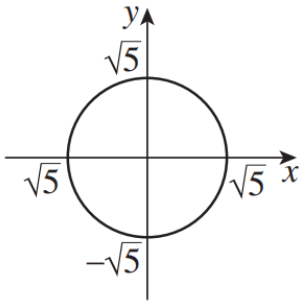
4 Determine the equation of each circle. They are all centred at the origin.

a



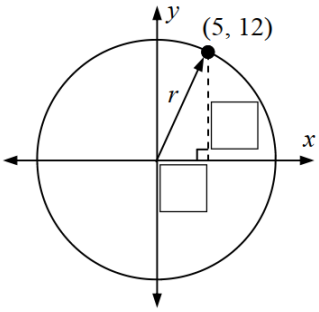
$x^2 + y^2 = 4$

b



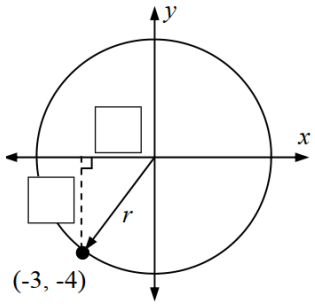
$x^2 + y^2 = 5$

c



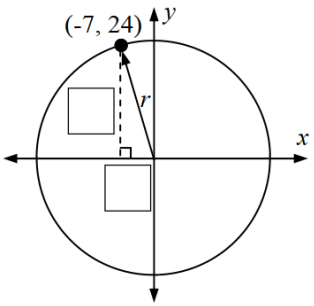
$x^2 + y^2 = 169$

d



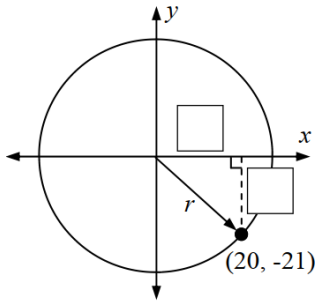
$x^2 + y^2 = 25$

e



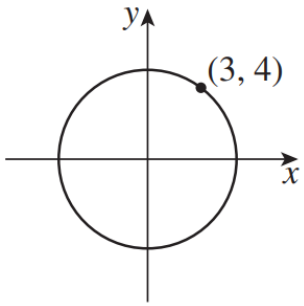
$x^2 + y^2 = 625$

f



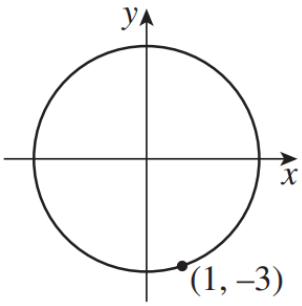
$x^2 + y^2 = 841$

g



$x^2 + y^2 = 25$

h



$x^2 + y^2 = 10$

5 A circle with centre $O(0,0)$ passes through $A(5,12)$.

a What is its radius?

13

b Therefore, what is its equation?

$x^2 + y^2 = 169$

6 Consider the four points $A(1,4), B\left(2,\sqrt{13}\right), C\left(3,2\sqrt{2}\right)$ and $D(4,1)$

a Find the distance of each point from the origin O.

$\sqrt{17}$

b Hence explain why the four points lie on a circle with centre the origin and state the equation of the circle.

Each point is equidistant from the origin, so it lies on a circle centred at the origin.

$x^2 + y^2 = 17$

c What is the radius, diameter, circumference and area of this circle?

$r = \sqrt{17}, d = 2\sqrt{17}, C = 2\pi\sqrt{17}, A = 17\pi$

7 Show, using substitution, that $E(-12,-5)$ lies on the circle centred at the origin with radius 13.

$x^2 + y^2 = 169$

$(-12)^2 + (-5)^2 = 144 + 25 = 169$

8 Consider points $A(-5,0), B(5,0)$, and $C(3,4)$.

a Show that the points all lie on the circle $x^2 + y^2 = 25$.

Using distance formula, distance from origin of all 3 points is 5

b Show that $AC \perp BC$.

$m_{AC} = \frac{1}{2}, m_{BC} = -2$

Since the gradients of AC and BC are negative reciprocals, they are perpendicular.

Circle Transformations

ⓘ General Equation of a Circle

centre (a, b)

$$(x-a)^2+(y-b)^2=r^2$$

radius r

🔍 **Identify** Centre and radius from equation of a circle

$(x-3)^2+(y-4)^2=25$

Centre $(3,4)$, Radius 5

$(x-4)^2+(y+3)^2=36$

Centre $(4,-3)$, Radius 6

Identify the centre and radius of these circles

a $(x-3)^2+(y+4)^2=16$

Centre ..... Radius .....

b $(x+3)^2+y^2=25$

Centre ..... Radius .....

c $(x+2)^2+(y-50)^2=100$

Centre ..... Radius .....

d $(x+19)^2+\left(y+\frac{1}{2}\right)^2=50$

Centre ..... Radius .....

Find the equation of the circle for:

e Centre: $(3,9)$, radius 4

f Centre: $\left(-1,\frac{3}{4}\right)$, radius $\sqrt{13}$

ⓘ Sketching a Translated Circle

centre (a, b)

$$(x-a)^2+(y-b)^2=r^2$$

radius r

1. Identify centre and radius.
2. Draw a quick sketch, using the radius to determine if it crosses the axes.
3. Find x and y intercepts, if any.

Example Sketch the graph of a circle not centred at the origin

Sketch $(x-1)^2+(y+2)^2=9$.

Centre $(1,-2)$ radius 3.

The circle will cross the axes because the radius is 3.

x intercepts let $y=0$

$(x-1)^2+(0+2)^2=9$

$(x-1)^2=5$

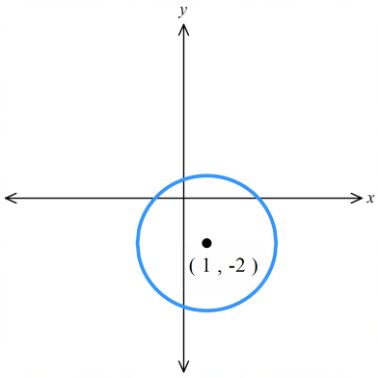
$x=1\pm\sqrt{5}$

y intercepts let $x=0$

$(0-1)^2+(y+2)^2=9$

$(y+2)^2=8$

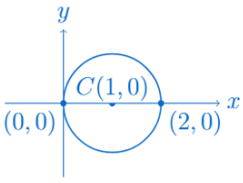
$y=-2\pm 2\sqrt{2}$



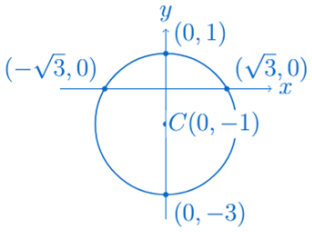
Guided Practice

Sketch, and find the intercepts, if any.

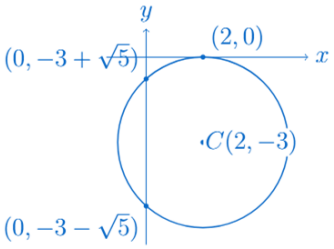
a $(x-1)^2+y^2=1$



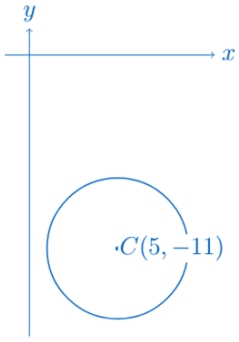
b $x^2+(y+1)^2=4$



c $(x-2)^2+(y+3)^2=9$



d $(x-5)^2+(y+11)^2=16$



1 Choose the word *left, right, up* or *down* to complete each sentence.

a Compared to $x^2 + y^2 = r^2$, $(x-1)^2 + y^2 = r^2$ is translated ..... by 1 unit.

right

b Compared to $x^2 + y^2 = r^2$, $(x+3)^2 + y^2 = r^2$ is translated .....

Left by 3 units

c Compared to $x^2 + y^2 = r^2$, $x^2 + (y+3)^2 = r^2$ is translated .....

Down by 3 units

d Compared to $x^2 + y^2 = r^2$, $x^2 + (y-2)^2 = r^2$ is translated .....

Up by 2 units

e Compared to $x^2 + y^2 = r^2$, $(x+2)^2 + (y-1)^2 = r^2$ is translated

.....

Left by 2, up by 1

f Compared to $x^2 + y^2 = r^2$, $(x+5)^2 + (y+3)^2 = r^2$ is translated

.....

Left by 5, down by 3

g Compared to $x^2 + y^2 = r^2$, $(x-4)^2 + (y+2)^2 = r^2$ is translated

.....

Right by 4, down by 2

2 Write the equation of the circle with:

a centre $(-1,5)$, radius 4

$(x+1)^2 + (y-5)^2 = 16$

b centre $(3,-5)$, radius $\sqrt{7}$

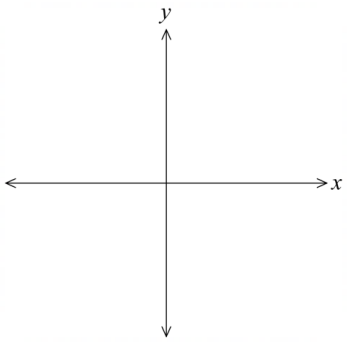
$(x-3)^2 + (y+5)^2 = 7$

c $(8,11)$, radius $2\sqrt{5}$

$(x-8)^2 + (y-11)^2 = 20$

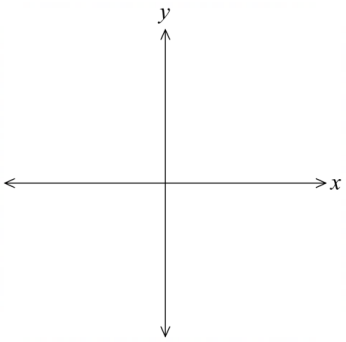
3 Sketch the following circles.
Label the coordinates of the centre and intercepts, if any. Check your sketch using DESMOS.

a $(x-1)^2+y^2=1$



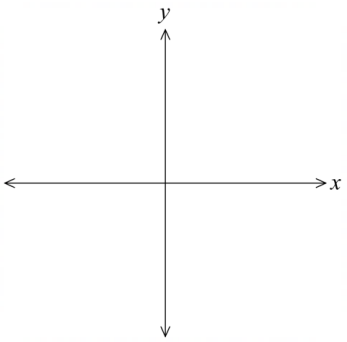
Centre: (1,0)
x-intercepts:$x=0,2$
y-intercepts:$y=0$

b $(x+1)^2+y^2=9$



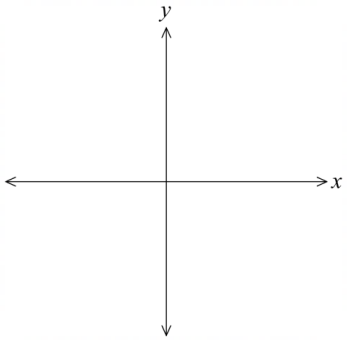
Centre: (−1,0)
x-intercepts:$x=-4,2$
y-intercepts:$y=\pm2\sqrt{2}$

c $x^2+(y-4)^2=36$



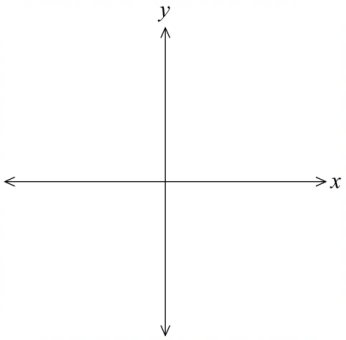
Centre: (0,4)
x-intercepts:$x=\pm2\sqrt{5}$
y-intercepts: $-2,10$

d $(x-3)^2+(y+1)^2=1$



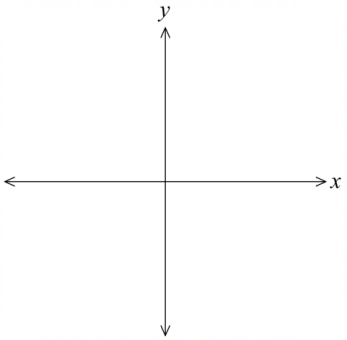
Centre: (3,−1)
x-intercepts:$x=3$
y-intercepts: none

e $(x+2)^2+(y-3)^2=4$



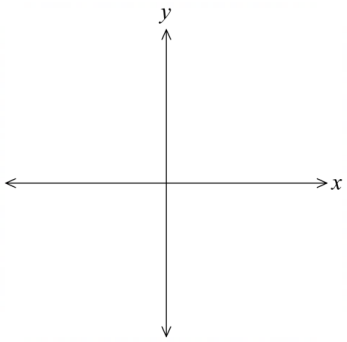
Centre: (−2,3)
x-intercepts: none
y-intercepts:3

f $(x-1)^2+(y+3)^2=25$



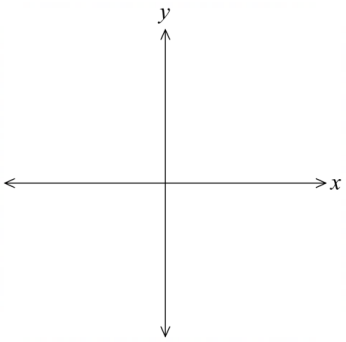
Centre: (1,−3)
x-intercepts:$x=-3,5$
y-intercepts:$y=-3\pm2\sqrt{6}$

g $(x+3)^2+(y-2)^2=25$



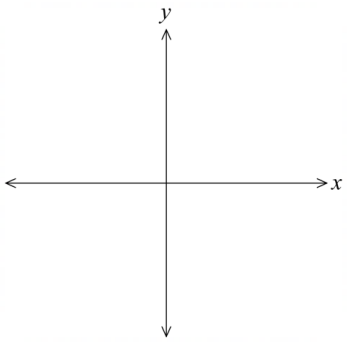
Centre: $(-3,2)$
x-intercepts: $x=-3\pm\sqrt{21}$
y-intercepts: $y=-2,6$

h $(x+2)^2+(y-1)^2=9$



Centre: $(-2,1)$
x-intercepts: $x=-2\pm2\sqrt{2}$
y-intercepts: $y=1\pm\sqrt{5}$

i $(x-2)^2+(y-5)^2=64$



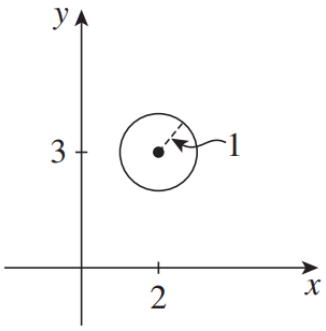
Centre: $(2,5)$
x-intercepts: $x=2\pm\sqrt{39}$
y-intercepts: $y=5\pm2\sqrt{15}$

4 A circle with centre $B(4,5)$ passes through the origin. What is its equation?

$(x-4)^2+(y-5)^2=41$

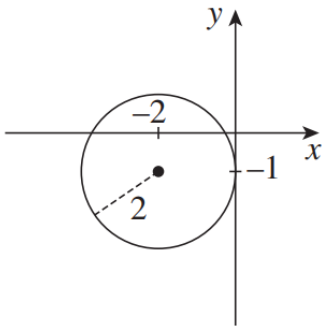
5 Determine the equation of each circle, using Pythagoras if necessary to find the radius.

a



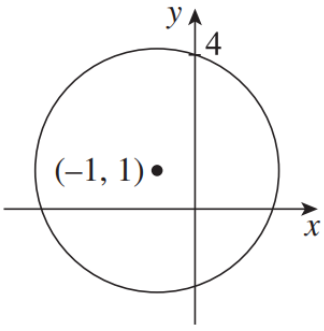
$(x-2)^2+(y-3)^2=1$

b



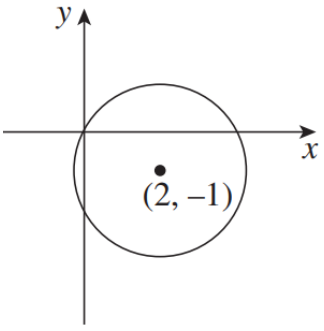
$(x+2)^2+(y+1)^2=4$

c



$(x+1)^2+(y-1)^2=10$

d



$(x-2)^2+(y+1)^2=5$

6 Find the equation of a circle with centre $(5, -2)$ passing through $(-1,1)$.

Distance between the centre and the point is $\sqrt{45}$, so the radius is $\sqrt{45}$.

$(x-5)^2 + (y+2)^2 = 45$

7 A circle has centre $(4, -3)$ and one of the x-intercepts is at $(-\sqrt{7} + 4, 0)$.
Determine the equation of the circle.

Distance between the centre and the intercept is 4, so the radius is 4.

$(x-4)^2 + (y+3)^2 = 16$

8 The interval joining the points $A(3, -1)$ and $B(-7,5)$ is a diameter of a circle.

a Find the midpoint of AB.

$(-2,2)$

b Explain why the midpoint of AB is the centre of the circle.

The diameter connects two points on the circumference, passing through the centre.

Therefore, the midpoint of the diameter is the centre of the circle.

c Find the length of the radius.

$\sqrt{34}$

d Hence state the equation of the circle.

$(x+2)^2 + (y-2)^2 = 34$

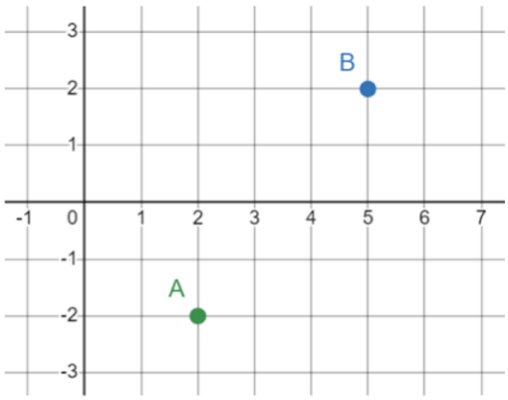
9 Find the centre of the circle with diameter CD, where $C = (2, 1)$ and $D = (8, -7)$.

$(5, -3)$

10 Find the equation of a circle with diameter KL, where $K(5,7)$ and $L(-9,3)$

$(x + 2)^2 + (y - 5)^2 = 53$

11 Find the possible equations of a circle with radius AB.



Circle with centre at A:

$(x - 2)^2 + (y + 2)^2 = 25$

Circle with centre at B:

$(x - 5)^2 + (y - 2)^2 = 25$

Completing the Square for a Circle

Completing the Square to Find the Centre and Radius

Some questions require you to **complete the square** for the x and y terms to write in the form

$$(x-a)^2+(y-b)^2=r^2$$

Example Sketch the graph of a circle by completing the square

$$x^2+2x+y^2-6y-6=0$$

$$x^2+2x+\mathbf{1}-\mathbf{1}+y^2-6y+\mathbf{9}-\mathbf{9}-6=0$$

$$(x^2+2x+1)-1+(y^2-6y+9)-9-6=0$$

$$(x+1)^2+(y-3)^2-16=0$$

$$(x+1)^2+(y-3)^2=16$$

Centre: $(-1,3)$, radius 4

Guided Practice

Find the centre and radius by completing the square first.

a $x^2-4x+y^2=0$

$(2,0)$ $r=2$

b $x^2+6x+y^2-8y=0$

$(-3,4)$ $r=5$

c $x^2+y^2-10x+12y=12=0$

$(5,-6)$ $r=7$

d $4x^2+4y^2-12x+20y-7=0$

$\left(\frac{3}{2},-\frac{5}{2}\right)$ $r=\frac{\sqrt{41}}{2}$

1 Determine the centre and radius of these circles by completing the square, if necessary.

a $(x+1)^2+y^2=4$

Centre: $(-1,0)$ Radius: 2

b $(x-1)^2+(y-2)^2=1$

Centre: $(1,2)$ Radius: 1

c $x^2-10x+y^2=-16$

Centre: $(5,0)$ Radius: 3

d $x^2+y^2+14x=33$

Centre: $(-7,0)$ Radius: $\sqrt{82}$

e $x^2-2x+y^2-4y-4=0$

Centre: $(1,2)$ Radius: 3

f $x^2+6x+y^2-8y=0$

Centre: $(-3,4)$ Radius: 5

g $x^2-10x+y^2+8y+32=0$

Centre: $(5,-4)$ Radius: 3

h $x^2+14x+14+y^2-2y=0$

Centre: $(-7,1)$ Radius: 6

i $x^2+4x+y^2-2y+1=0$

Centre: $(-2,1)$ Radius: 2

j $x^2-6x+y^2-4y-3=0$

Centre: $(3,2)$ Radius: 4

2

a Complete the square for $x^2+4x+y^2-6y+15=0$.

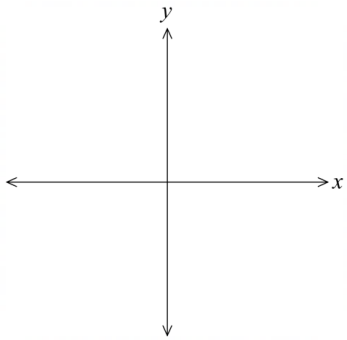
$(x+2)^2+(y-3)^2=-2$

b Explain why this is not an equation of a circle.

The radius cannot be negative.

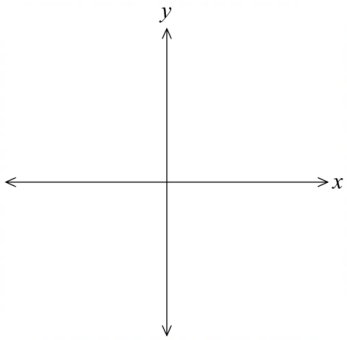
3 Complete the square to sketch these circles. Show the centre, radius, and intercepts.

a $x^2 - 2x + y^2 = 0$



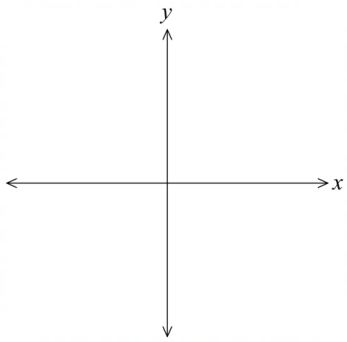
Centre: (1,0)
Radius: 1
x-intercepts: $x = 0, 2$
y-intercepts: $y = 0$

b $x^2 + 6x + y^2 = 0$



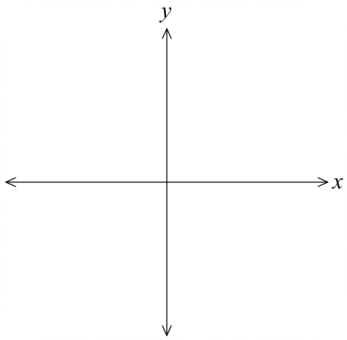
Centre: (–3,0)
Radius: 3
x-intercepts: $x = -6, 0$
y-intercepts: $y = 0$

c $x^2 + 4x + y^2 - 2y + 1 = 0$



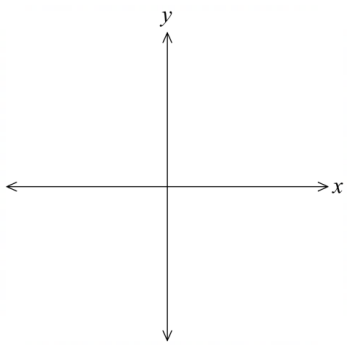
Centre: (–2,1)
Radius: 2
x-intercepts: $x = -2 \pm \sqrt{3}$
y-intercepts: $y = 1$

d $x^2 + 8x + y^2 + 10y + 5 = 0$



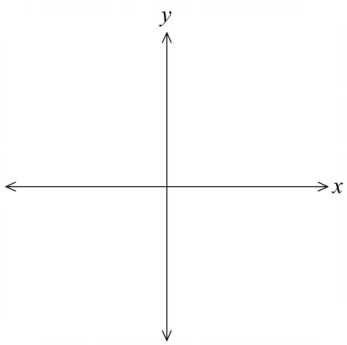
Centre: (–4,–5)
Radius: 6
x-intercepts: $x = -4 \pm \sqrt{11}$
y-intercepts: $y = -5 \pm 2\sqrt{5}$

e $x^2 - 6x + y^2 - 4y - 3 = 0$



Centre: (3,2)
Radius: 4
x-intercepts: $x = 3 \pm 2\sqrt{3}$
y-intercepts: $y = 2 \pm \sqrt{7}$

f $x^2 - 2x + y^2 + 6y - 5 = 0$



Centre: (1,–3)
Radius: $\sqrt{15}$
x-intercepts: $x = 1 \pm \sqrt{6}$
y-intercepts: $y = -3 \pm \sqrt{14}$

4 For what values of k would the graph of $x^2 + 6x + y^2 + k = 0$ not exist?

By CTS, $(x+3)^2 + y^2 = 9 - k$

For circle to not exist, $9 - k < 0$

$\therefore k > 9$

5 A naval ship is at coordinates P and equipped with a radar scanner that can detect objects inside the circle $x^2 + y^2 - 80x - 40y = 500$, where radius is measured in kilometres.

a Find the coordinates of P.

By CTS, $(x-40)^2 + (y-20)^2 = 2500$

P is the centre of the circle, at $(40,20)$

b Another ship is at $Q(60, -20)$ and can detect objects within a radius of 40 km.

i Will ship Q be able to detect P?

Distance between PQ is $\sqrt{2000} \approx 44.7$ km. Since Q has a radar range of 40 km, no.

ii Will ship P be able to detect Q?

P's radar has a range of 50 km, so yes.

6 Consider the points $A(1,3)$, $B(3,9)$ and let $P(x,y)$ be a point.

a Find an expression for the gradients of PA and PB.

$$m_{PA} = \frac{y-3}{x-1}, \quad m_{PB} = \frac{y-9}{x-3}$$

b Suppose $\angle APB = 90°$. Show that $x^2 - 4x + y^2 - 12y + 30 = 0$.

$$\begin{aligned} m_{PA} \times m_{PB} &= -1 \\ \frac{y-3}{x-1} \times \frac{y-9}{x-3} &= -1 \\ \frac{(y-3)(y-9)}{(x-1)(x-3)} &= -1 \\ (y-3)(y-9) &= -(x-1)(x-3) \\ y^2 - 12y + 27 &= -x^2 + 4x - 3 \\ x^2 - 4x + y^2 - 12y + 30 &= 0 \end{aligned}$$

c By completing the square, show that this is the equation of a circle centred at $(2,6)$ with radius $\sqrt{10}$.

d How is the point P related to the circle? A sketch using DESMOS may help.
https://www.desmos.com/geometry/opfcwyddok

We have proved that if we draw any point P and connect it to both A and B such that it forms a right angle at $\angle APB$, the point P must lie on a circle centred at the midpoint of AB with radius $\sqrt{10}$.

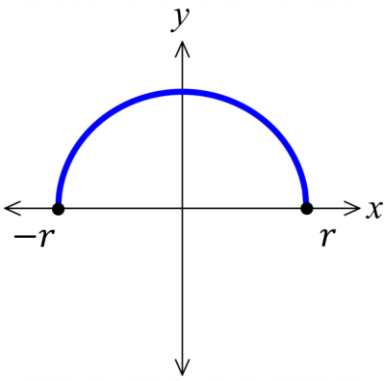
This is related to the circle theorem "The angle at the circumference in a semicircle is 90°"

Semicircles

Semicircle Functions

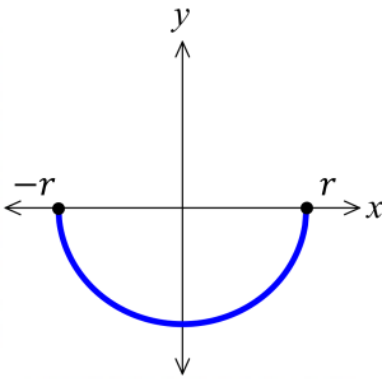
Positive semicircle function

$$y = \sqrt{r^2 - x^2}$$



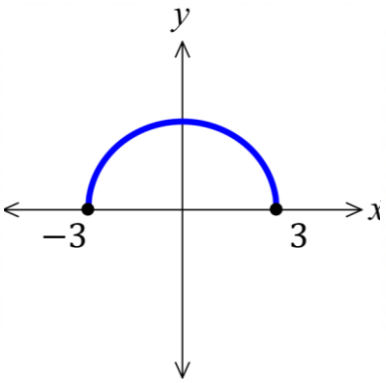
Negative semicircle function

$$y = -\sqrt{r^2 - x^2}$$

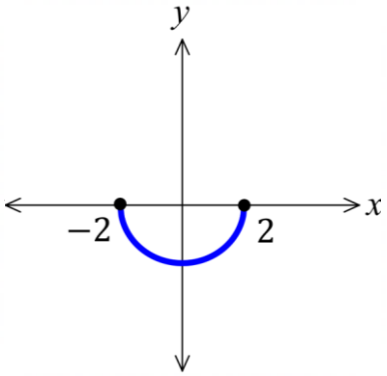


Example Sketch semicircles

$y = \sqrt{9 - x^2}$



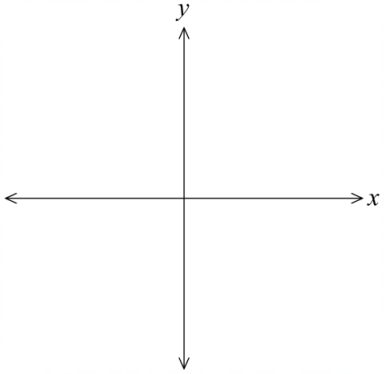
$y = -\sqrt{4 - x^2}$



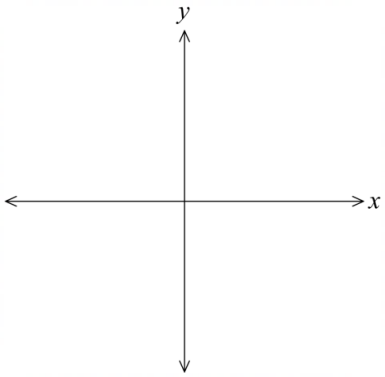
Guided Practice

Sketch

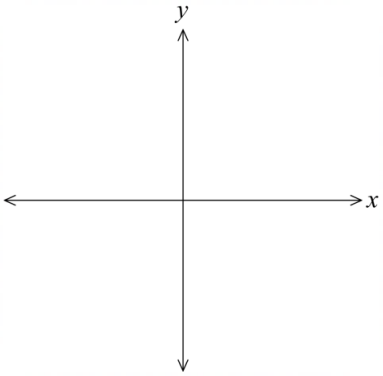
a $y = \sqrt{4 - x^2}$



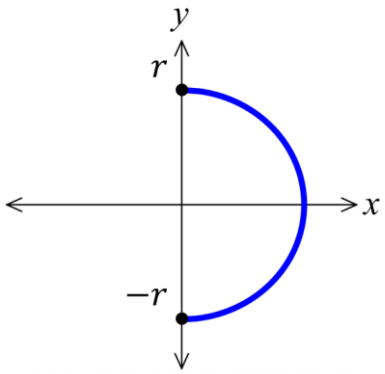
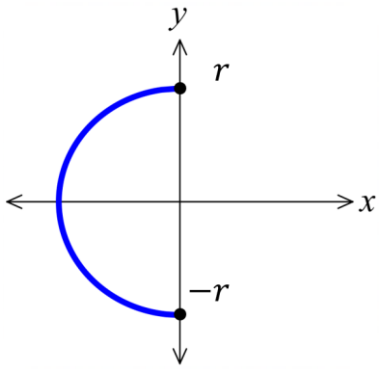
b $y = -\sqrt{8 - x^2}$



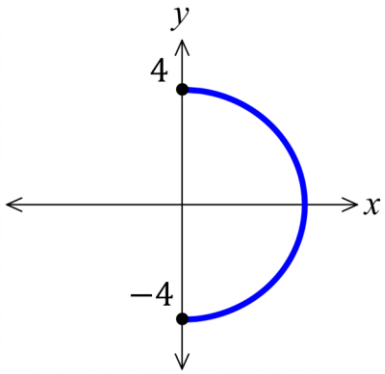
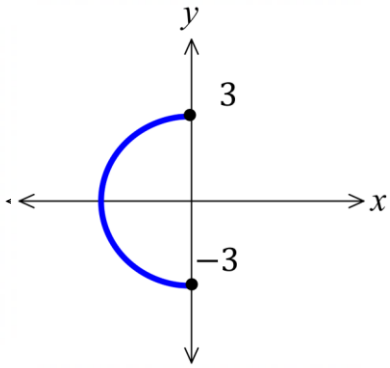
c $y = \sqrt{\frac{4}{9} - x^2}$



Semicircle Functions

Positive semicircle	Negative semicircle
$x = \sqrt{r^2 - y^2}$	$x = -\sqrt{r^2 - y^2}$
	

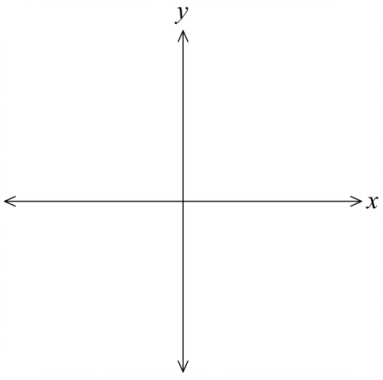
Example Sketch semicircles

$y = \sqrt{16 - x^2}$	$y = -\sqrt{9 - x^2}$
	

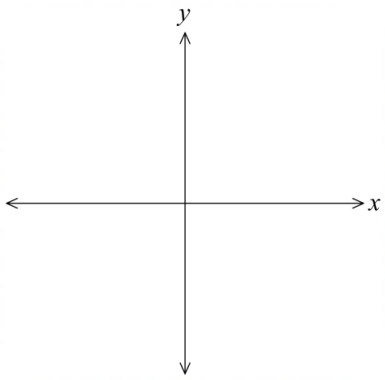
Guided Practice

Sketch

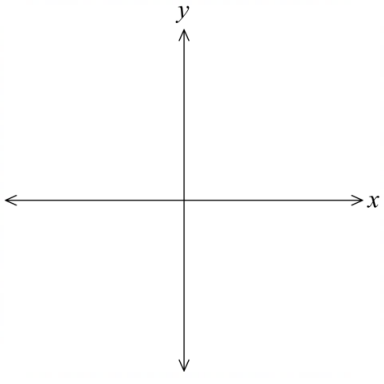
a $x = \sqrt{25 - y}$



b $x = -\sqrt{6 - x^2}$

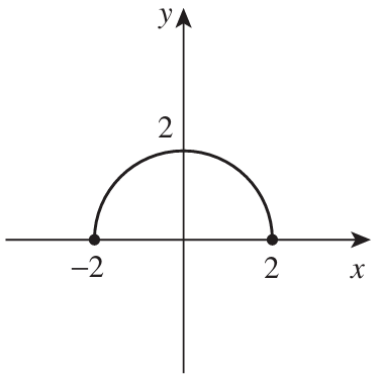


c $x = \sqrt{\frac{16}{9} - y^2}$



1 Sketch each semicircle. State the domain and range.

a $y = \sqrt{4 - x^2}$



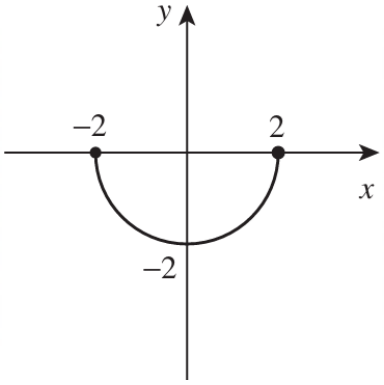
Domain:

[−2,2]

Range:

[0,2]

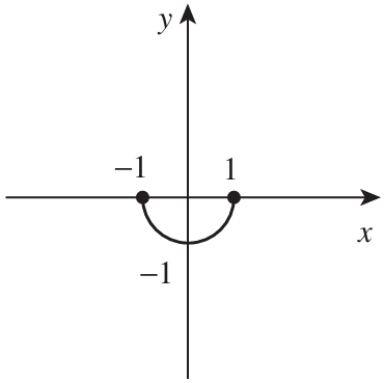
b $y = -\sqrt{4 - x^2}$



D: [−2,2]

R: [−2,0]

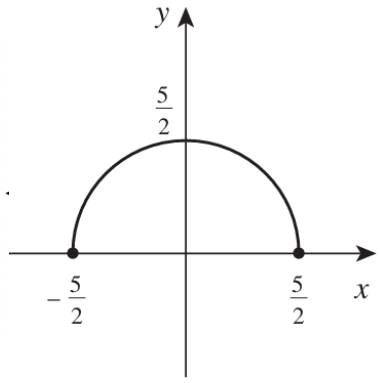
c $y = -\sqrt{1 - x^2}$



D: [−1.1]

R: [−1,0]

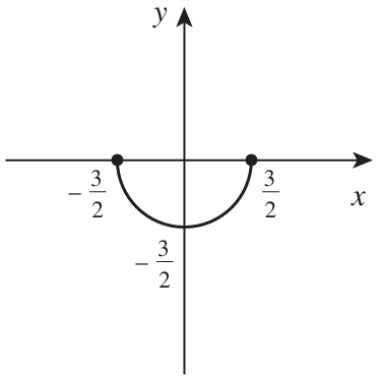
d $y = \sqrt{\frac{25}{4} - x^2}$



D: $\left[-\frac{5}{2}, \frac{5}{2}\right]$

R: $\left[0, \frac{5}{2}\right]$

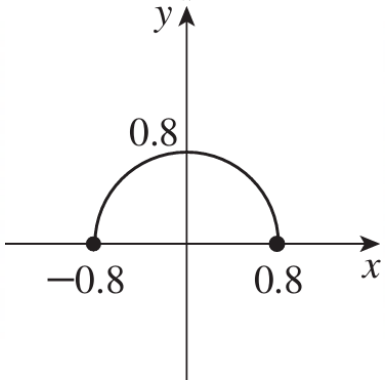
e $y = -\sqrt{\frac{9}{4} - x^2}$



D: $\left[-\frac{3}{2}, \frac{3}{2}\right]$

R: $\left[-\frac{3}{2}, 0\right]$

f $y = \sqrt{0.64 - x^2}$

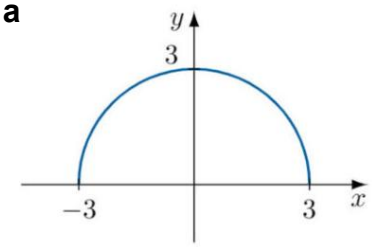


D: [−0.8,0.8]

R: [0,0.8]

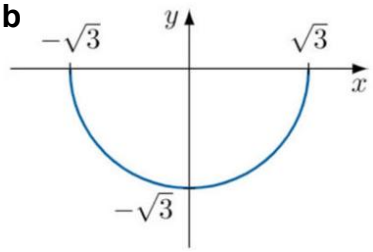
2 Write the equation of these semicircles.

a



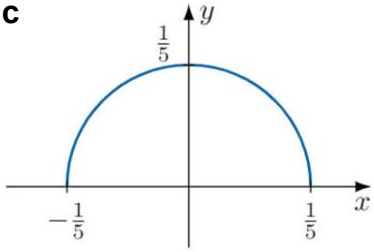
$y = \sqrt{9 - x^2}$

b



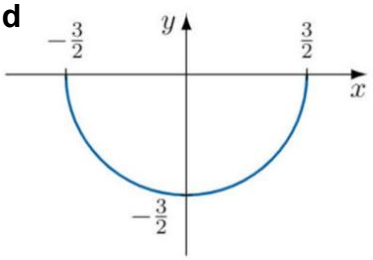
$y = -\sqrt{3 - x^2}$

c



$y = \sqrt{\frac{1}{25} - x^2}$

d

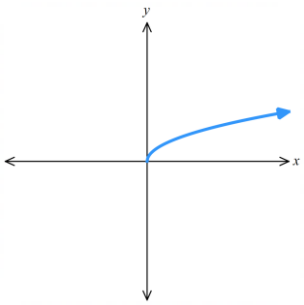
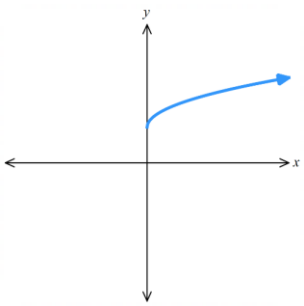
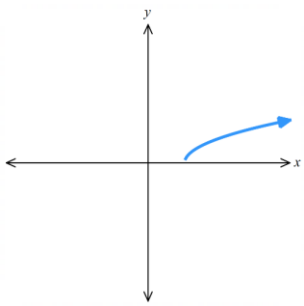
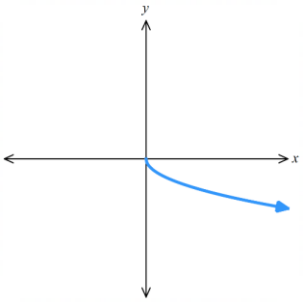


$y = -\sqrt{\frac{9}{4} - x^2}$

Square Root Functions

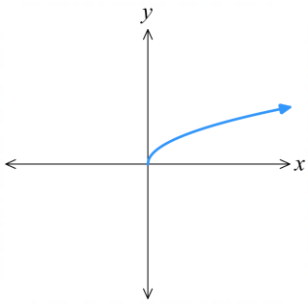
ⓘ Square Root Function

- The square root function $f(x) = \sqrt{x}$ is half a sideways parabola.
- Often questions will ask for the domain and range of a square root function; we can consider the equation or the graph.

$y = \sqrt{x}$	$y = \sqrt{x} + c$	$y = \sqrt{x - b}$	$y = -\sqrt{x}$
			

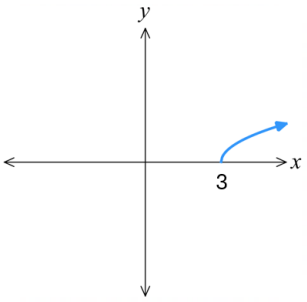
🔍 Domain and Range of Square Root Functions

a $f(x) = \sqrt{x}$



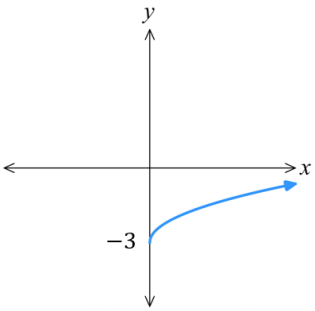
Domain $x \geq 0$ Range $y \geq 0$

b $f(x) = \sqrt{x - 3}$



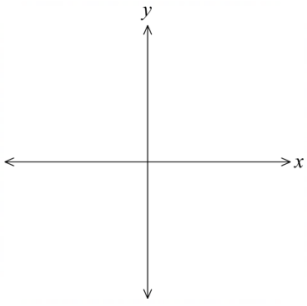
Domain $x \geq 3$ Range $y \geq 0$

c $f(x) = \sqrt{x} - 3$



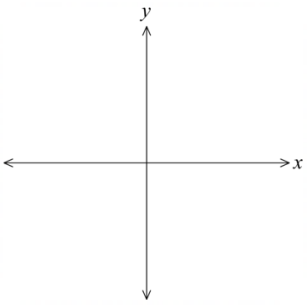
Domain $x \geq 0$ Range $y \geq -3$

d $f(x) = \sqrt{x} + 3$



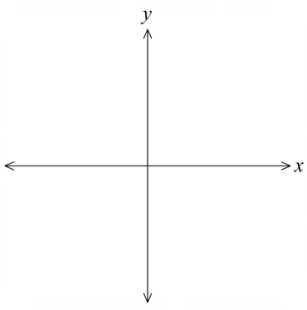
Domain $x \geq 0$ Range $y \geq 3$

e $f(x) = \sqrt{x + 3}$



Domain $x \geq -3$ Range $y \geq 0$

f $f(x) = -\sqrt{x + 3}$



Domain $x \geq -3$ Range $y \leq 0$

1 State the domain and range of these functions and sketch the function.

a $y = \sqrt{x}$	**b** $y = \sqrt{x-2}$	**c** $y = \sqrt{x-5}$
d $y = \sqrt{x+5}$	**e** $y = \sqrt{x+5}+1$	**f** $= \sqrt{x+5}-4$
g $y = -\sqrt{x+5}-4$	**h** $y = -\sqrt{x+5}$	**i** $y = 2\sqrt{x+5}$
j $y = \sqrt{2x+10}$	**k** $y = \sqrt{2x-10}$	**l** $y = \sqrt{-x}$
m $y = \sqrt{5-x}$	**n** $y = \sqrt{5-x}+3$	**o** $y = -\sqrt{5-x}+3$

Points of Intersection of Two Curves

ⓘ Finding the Points of Intersection of Any Two Graphs

For any two graphs,

1. Solve simultaneously to find the x-coordinate(s) of the point of intersection.

2. Find the y-coordinate(s).

Example Find points of intersection of a line and a hyperbola

Find the points of intersection of the line $y = 4x$ and the hyperbola $y = \frac{1}{x}$

1. Solve for x.

$4x = \frac{1}{x}$

$4x^2 = 1$

$x^2 = \frac{1}{4}$

$x = \pm\frac{1}{2}$

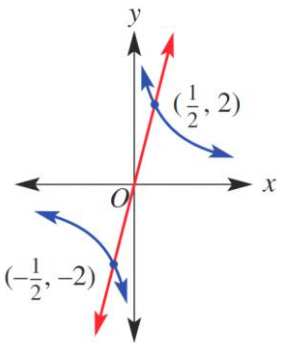
2. Find y values.

Substitute x into (1) to find y values:

When $x = \frac{1}{2}$ $\quad y = 4\left(\frac{1}{2}\right) = 2$

When $x = \frac{-1}{2}$ $\quad y = 4\left(\frac{-1}{2}\right) = -2$

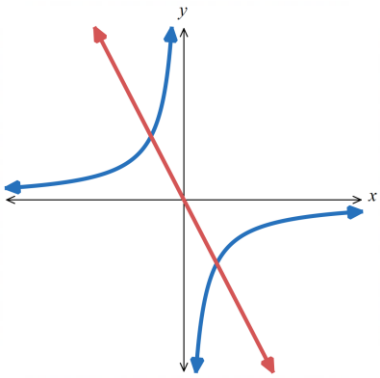
$\therefore \left(\frac{1}{2}, 2\right)$ and $\left(-\frac{1}{2}, -2\right)$



Guided Practice

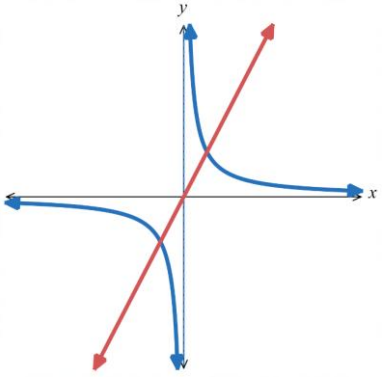
Find the point of intersection of these graphs.

a $y = -\frac{2}{x}$ and $y = -2x$



$(-1,2)$ and $(1,-2)$

b $y = \frac{1}{x}$ and $y = 2x$



$\left(-\frac{1}{\sqrt{2}}, -\sqrt{2}\right)$ and $\left(\frac{1}{\sqrt{2}}, \sqrt{2}\right)$

Example Find points of intersection of a line and a circle

Find the points of intersection of the line $y = 2x$ and the circle $y^2 + x^2 = 4$

1. Solve simultaneously

$y = 2x$ (1)

$y^2 + x^2 = 4$ (2)

Sub (1) into (2)

$(2x)^2 + x^2 = 4$

$5x^2 = 4$

$x^2 = \frac{4}{5}$

$x = \pm\frac{2}{\sqrt{5}}$

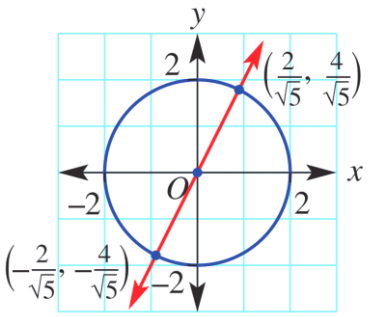
3. Find y values.

Substitute into (1) to find y values:

When $x = \frac{2}{\sqrt{5}}$ $\quad y = 2\left(\frac{2}{\sqrt{5}}\right) = \frac{4}{\sqrt{5}}$

When $x = \frac{-2}{\sqrt{5}}$ $\quad y = 2\left(\frac{-2}{\sqrt{5}}\right) = \frac{-4}{\sqrt{5}}$

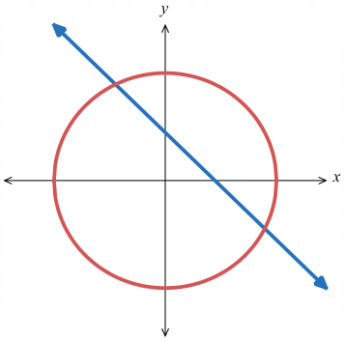
$\therefore \left(\frac{2}{\sqrt{5}}, \frac{4}{\sqrt{5}}\right)$ and $\left(\frac{-2}{\sqrt{5}}, \frac{-4}{\sqrt{5}}\right)$



Guided Practice

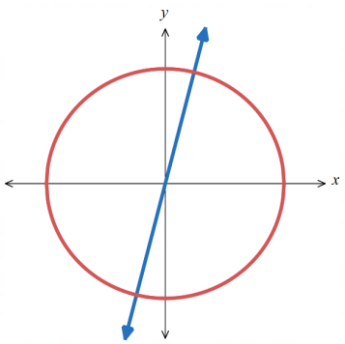
Find the point of intersection of these graphs.

a $y = -x + 1$ and
$x^2 + y^2 = 5$



$(-1,2)$ and $(2,-1)$

b $x^2 + y^2 = 16$ and
$y = 4x$



$\left(-\frac{4}{\sqrt{17}}, -\frac{16}{\sqrt{17}}\right)$ and $\left(\frac{4}{\sqrt{17}}, \frac{16}{\sqrt{17}}\right)$

1 Find the point of intersection of these graphs. You will need to rearrange before equating.

a $y = x^2$ and $x = \frac{y}{2}$

(0,0) and (2,4)

b $y + 4x^2 + x - 6 = 0$ and $y - 3x = 7$

(−0.5,5.5)

c $y = x^2 + 3x - 1$ and $3y = x - 3$

$\left(-\frac{8}{3}, -\frac{17}{9}\right)$ and $(0, -1)$

d $y - 7 = (x - 2)^2$ and $y + x = 9$

(1,8) and (2,7)

e $xy = 1$ and $y = x - 1$

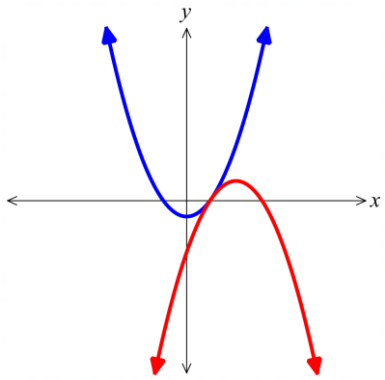
$\left(\frac{1-\sqrt{5}}{2}, \frac{-1-\sqrt{5}}{2}\right)$ and $\left(\frac{1+\sqrt{5}}{2}, \frac{\sqrt{5}-1}{2}\right)$

f $x^2 + y^2 = 4$ and $2y + x = 0$

$\left(\frac{-4}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$ and $\left(\frac{4}{\sqrt{5}}, -\frac{2}{\sqrt{5}}\right)$

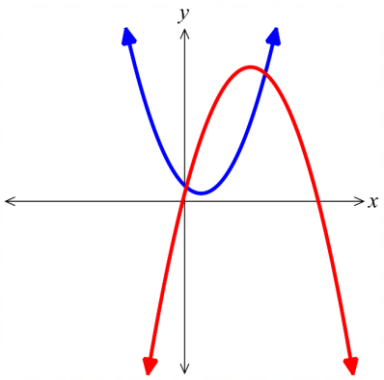
2 Find the point of intersection for these functions.
A sketch has been provided to help you interpret your answer.

a $y = x^2 - 2$
$y = -x^2 + 6x - 6.5$



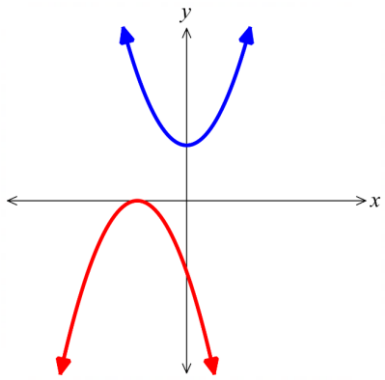
(1.5, 0.25)

b $y = x^2 - 2x + 2$
$y = -x^2 + 8x + 1$



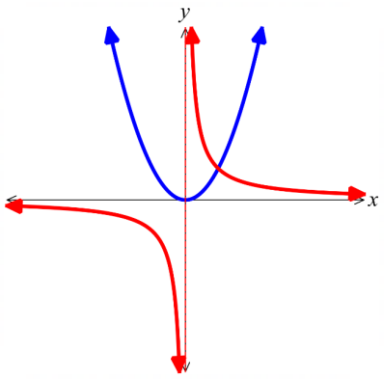
$\left(\frac{5-\sqrt{23}}{2}, 9-\frac{3\sqrt{23}}{2}\right)$ and $\left(\frac{5+\sqrt{23}}{2}, 9+\frac{3\sqrt{23}}{2}\right)$

c $y = x^2 + 7$
$y = -x^2 + 6x - 9$



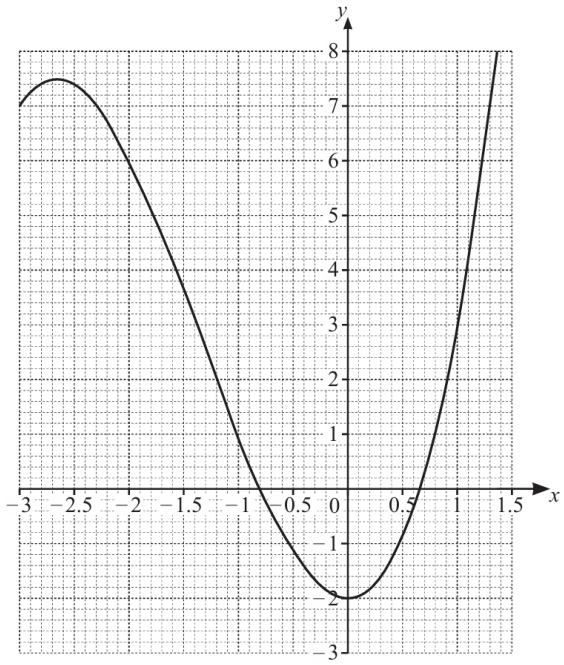
You will end up with a quadratic where $\Delta < 0$,
no points of intersection

d $y = x^2$
$y = \frac{8}{x}$



(2,4)

3 The diagram shows the graph of $y = x^3 + 4x^2 - 2$ for $-3 \leq x \leq 1.5$.
By drawing a suitable straight line, solve the equation $x^3 + 4x^2 - 2 = 2x$ for $-3 \leq x \leq 1.5$.
Give your answer as estimates to 2 d.p.



Sketch the line $y = 2x$ on the same axis.

The x-coordinates of the points of intersection are the solution.

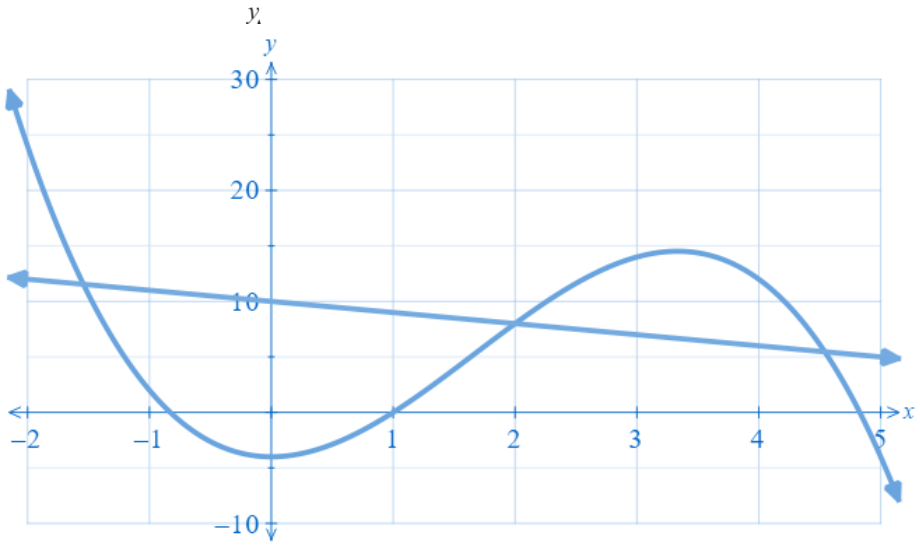
$x \approx -0.52$ and 0.88

4 The table shows some values for $y = 5x^2 - x^3 - 4$

x	-2	-1	0	1	2	3	4	5
y	24		-4		8	14		-4

a Complete the table.

b Sketch the graph for $-2 \leq x \leq 5$.



c By drawing a suitable straight line on the grid, solve the equation $x^3 - 5x^2 - x + 14 = 0$

This is equivalent to solving $5x^2 - x^3 - 4 = 10 - x$

Draw the line $y = 10 - x$ and read off the points of intersection

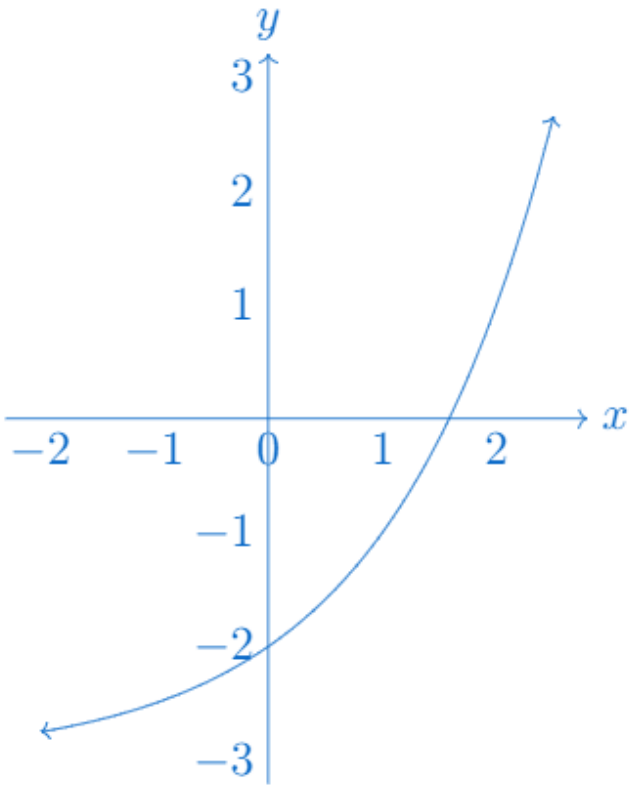
$x \approx -1, 5, 2, 4.5$

5 The table shows some values for $y = 2^x - 3$

a Complete the table.

x	-2	-1	0	0.5	1	1.5	2	2.5
y	-2.75			-1.58		-0.17	1	2.66

b On the grid, sketch the graph $y = 2^x - 3$ for $-2 \leq x \leq 2.5$.



c Use your graph to solve $2^x - 3 = 2$.

Draw a horizontal line $y = 2$ and read off points of intersection

$x \approx 2.3$

d By drawing a suitable straight line, solve $2^x - x - 1.5 = 0$.

This is equivalent to solving $2^x - 3 = x - 1.5$

Sketch $y = x - 1.5$ and read off x values of points of intersection

$x = -1$ and ≈ 1.66

6 Show that $y = -2x + 5$ is tangent to the circle $x^2 + y^2 = 5$.

Equate to find points of intersection:

$x^2 + (-2x + 5)^2 = 5$

$x^2 + 4x^2 - 20x + 25 = 5$

$5x^2 - 20x + 20 = 0$

$\Delta = b^2 - 4ac = (-20)^2 - 4(5)(20) = 0$

$\therefore$ there is 1 point of intersection, hence the line is tangent to the circle.

7 Show that $3x + y = 6$ is tangent to the circle $x^2 + y^2 + 4x + 16y + 28 = 0$.

Same method as above.

$x^2 + (-3x + 6)^2 + 4x + 16(-3x + 6) + 28 = 0$

This simplifies to: $x^2 - 8x + 16 = 0$

$\Delta = b^2 - 4ac = (-8)^2 - 4(1)(16) = 0$

8 The line $2x - y + 3 = 0$ intersects the circle $x^2 + y^2 - 4x + 6y - 12 = 0$ at two points, D and E.

a Find the coordinates of D and E.

Rearrange equation of line: $y = 2x + 3$

Sub into equation of circle

$x^2 + (2x + 3)^2 - 4x + 6(2x + 3) - 12 = 0$

This simplifies to $5x^2 + 20x + 15 = 0$

$\therefore (x + 1)(x + 3) = 0$

The points of intersection are $(-1, 1), (-3, -3)$

b Hence find the length of the line segment joining DE.

$DE = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$= \sqrt{2^2 + 4^2} = 2\sqrt{5}$ units

9 For the circle $x^2 + y^2 = 4$ and line $y = mx + 4$, determine the exact values of the gradient, m, so that the line:

a is a tangent to the circle (intersects the circle once)

set up equation $x^2 + (mx + 4)^2 = 4$

$\Delta = 16m^2 - 48$

Let $\Delta = 0$

$m = \pm\sqrt{3}$

b intersects the circle at two points

$\Delta > 0$

$16m^2 - 48 > 0$

$m < -\sqrt{3}$ or $m > \sqrt{3}$

c does not intersect the circle

$-\sqrt{3} < m < \sqrt{3}$

10

a Try to find the coordinates of the points of intersection of $y = \frac{1}{x}$ and $y = -x + 1$.

What do you notice?

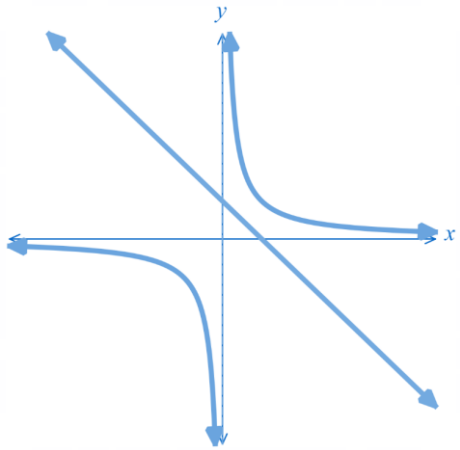
$-x + 1 = \frac{1}{x}$

$-x^2 + x = 1$

$x^2 - x + 1 = 0$

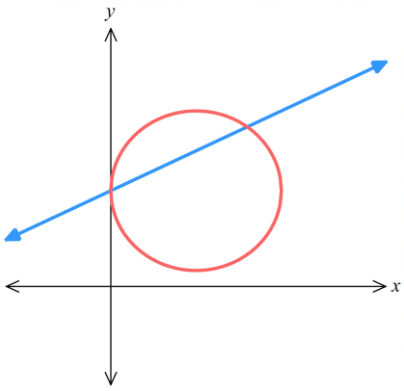
There are no points of intersection; $\Delta = (-1)^2 - 4(1)(1) = -3 < 0$.

b Sketch the two graphs on the same axis to visualise the scenario.



11 The line $2y - x = 12$ intersects the circle $x^2 + y^2 - 10x - 12y + 36 = 0$ at the points A and B.
A sketch is given, it would be useful to add to it as you go through parts a to d.

a Find the coordinates of A and B.



Using same method as previous question,

$(0,6)$ and $(8,10)$

b Find the equation of the perpendicular bisector of AB.

The perpendicular bisector of AB is the perpendicular line passing through the midpoint of AB.

Midpoint of AB: $\left(\frac{0+8}{2}, \frac{6+10}{2}\right) = (4,8)$

Gradient of AB: $m = \frac{10-6}{8-0} = \frac{1}{2}$

$\therefore$ gradient of perpendicular bisector is -2

Using $m = -2$ and $(4,8)$: $y - y_1 = m(x - x_1)$

$y - 8 = -2(x - 4)$

$y = -2x + 16$

c The perpendicular bisector of AB intersects the circle at the points P and Q.
Find the exact coordinates of P and Q.

Using same method as part **a**, $(5 + \sqrt{5}, 6 - 2\sqrt{5})$ and $(5 - \sqrt{5}, 6 + 2\sqrt{5})$

d Explain why APBQ is a kite and hence find its area.

The diagonals meet at right angles, and $PA = PB$ and $QA = QB$, so it satisfies the definition of a kite.

Using distance formula:

Length of AB is: $4\sqrt{5}$

Length of PQ is: 10

$A = \frac{xy}{2} = \frac{4\sqrt{5}(10)}{2} = 20\sqrt{5}$

12 Mr Ding used ChatGPT to generate an exam question:
'Find the points of intersection between $y = 2^x$ and $y = x^2 + 1$'
He checked on DESMOS and saw two 'nice' points of intersection at $(0,1)$ and $(1,2)$.
However, his students would not be able to find these points algebraically. Explain why.

It is not possible to solve $2^x = x^2 + 1$ using any standard algebraic techniques.

13 [Auxiliary circle of a hyperbola]
The equation $(x + y)^2 + (x - y)^2 = a^2$ is an equation representing a circle.

a By expanding then completing the square, find the radius of a circle in this form, in terms of a.

$r = \frac{a}{\sqrt{2}}$

b What kind of curve is $(x + y)^2 - (x - y)^2 = a^2$? (hint: expand and simplify)

Hyperbola with equation $y = \frac{a^2}{4x}$

c Use DESMOS to sketch $(x + y)^2 + (x - y)^2 = a^2$ and $(x + y)^2 - (x - y)^2 = a^2$.
Create a slider for a and experiment with values of a.
https://www.desmos.com/calculator/luwgidnzzy
What do you think the coordinates of the point of intersection are?

The points of intersection are $\left(\pm\frac{a}{2}, \pm\frac{a}{2}\right)$

d Verify this algebraically.

Start by expanding out each equation, after fully simplifying you should have $2x^2 + 2y^2 = a^2$ and $4xy = a^2$

Substitute $y = \frac{a^2}{4x}$ into the first equation to get $32x^4 - 16a^2x^2 + 2a^4 = 0$

Divide by the common factor and recognise as a perfect square: $(4x^2 - a^2)^2 = 0$

Solve to get $x = \pm\frac{a}{2}$

Substitute into a previous equation and solve for y: $y = \pm\frac{a}{2}$